Recommended Assessment

Force Resistor

Exploring Force Resistor Inputs

1. Write an equation for the voltage V_F across the variable resistor with resistance R_F. What is the V_F when $R_F = 0\Omega$, $R_F = 100k\Omega$, and $R_F = 1000k\Omega$? What is the V_F when $R_F = 100k\Omega$?

2. Using your multimeter measurements, what happens to the resistance when you apply force?

3. What happens to the measured voltage as you press the force resistor between varying surfaces and at different areas on the membrane? What happens when you reverse the terminals in the connection?

4. Refer to the datasheet and report the force range for the force resistor. Why is this an important consideration when selecting a force resistor?

5. Refer to the circuit diagram in Figure 3 of the datasheet. How does it differ from the way that the force resistor is connected to read its input using the sensors trainer? How does this affect the measured voltage across the force resistor?

Exploring Force Resistor Control

6. Consider the use of a force sensing resistor to turn on an LED when a minimum amount of force has been applied. What is the trade-off to consider when increasing or decreasing the force threshold?

7. Consider the following plot showing how resistance changes with applied force in a typical force sensing resistor (not the exact resistor that you are using for this lab). Compared to a linear force-resistance relationship, why is it difficult to obtain an exact value for the applied force using a force sensing resistor where the relationship is non-linear?

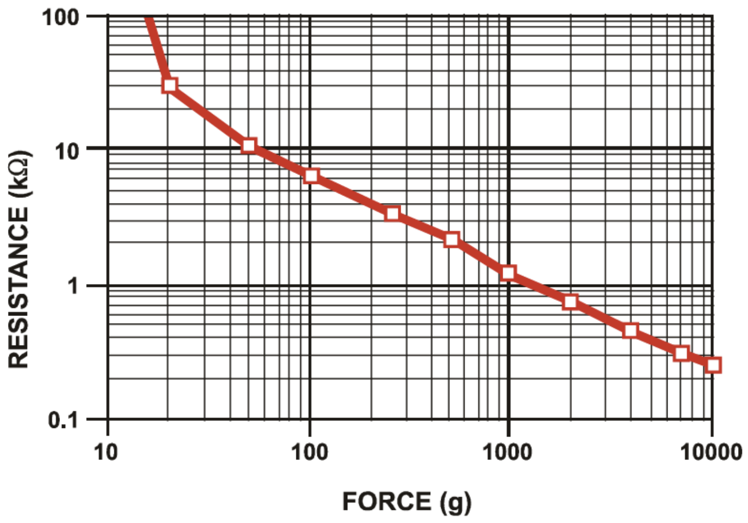

Figure from: https://cdn.sparkfun.com/assets/e/3/b/3/8/force_sensing_resistor_guide.pdf

Force Resistor Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	